Florence **BANSEPT**

ECOLOGY & EVOLUTION OF HOST-ASSOCIATED MICROBIOTA

Laboratoire de Chimie Bactérienne
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ACADEMIC PATH

2023 - ...
GROUP LEADER – *Laboratoire de Chimie Bactérienne, Marseille, France*
Since 11/2024: Chargée de Recherche at CNRS (permanent researcher)
Fellow of the Turing Center for Living Systems (CENTURI, AMU)
Permanent Visitor at the Center for Theoretical Physics (CPT, AMU)
Max Planck Partner Group of the Max Planck Institute for Evolutionary Biology (Plön, Germany)

2019 - 2022
POSTDOC - *Max Planck Institute for Evolutionary Biology, Plön, Germany*
Advisor: Arne TRAULSEN - Department of Theoretical Biology
Project: ***Modeling of microbial communities***

2015 - 2018
PHD THESIS - *Laboratoire Jean Perrin, Sorbonne Université, Paris, France*
Supervisors: Raphaël VOITURIEZ & Claude LOVERDO
Dissertation title: ***"Biophysical modeling of bacterial population dynamics & the immune response in the gut"***
Defense date: 05-12-2018
Funding: 3 years full fellowship (EDPIF) & teaching assistantship (UPMC)
Jury: Aleksandra WALCZAK, DR, *LPTENS Paris* – president
Bahram HOUCHMANDZADEH, DR, *LIPHY Grenoble* – reporter
Agnese SEMINARA, CR, *LPMC Nice* – reporter
Silvia DE MONTE, CR, *IBENS Paris* – examiner
Andrea PARMEGGIANI, PR, *L2C Montpellier* – examiner
Raphaël VOITURIEZ, DR, *LJP Paris* – supervisor
Claude LOVERDO, CR, *LJP Paris* – invited member
Emma WETTER-SLACK, Ass. Prof., *ETH Zurich* – invited member

2012 - 2015
ENS DIPLOMA - *École Normale Supérieure, Paris, France*
Major Physics - minor Musicology

2013 - 2015
MASTER "Macroscopic physics and Complexity" - with honors
École Normale Supérieure, International Center for Fundamental Physics, Paris, France

2nd year research internship (6 months) - *Laboratoire Jean Perrin, UPMC, Paris, France*
Supervisor: Claude LOVERDO
Project: ***Colonization dynamics of* Salmonella *Typhimurium in mice***

1st year research internship (5 months) - *Institut de Physique, UFRGS, Porto Alegre, Brazil*
Supervisor: Jeferson ARENZON
Project: ***Bootstrap percolation: Fast algorithm and applications***

2012 - 2013
BACHELOR in Fundamental Physics – *École Normale Supérieure, Paris, France*

Experimental internship (1 month) – *EC2M, Gulliver, ESPCI, Paris, France*
Supervisor: Olivier DAUCHOT
Project: ***Vibrated self-propelled particles under flexible confinement***

2010 - 2012
CLASSES PRÉPARATOIRES aux Grandes Écoles – *Lycée Kléber, Strasbourg, France*
Intensive undergraduate training in mathematics and physics to join top engineering schools

2010
BACCALAURÉAT Série S - with first class honors – *Lycée Lambert, Mulhouse, France*
A-Levels in mathematics and physics

Scientific output

Peer reviewed publications in international scientific journals *Nature, PLOS Comp Bio, Evolution, Trends in Microbio, ISME-J, BMC EcoEvo, eLife*	10
Preprints and manuscripts in preparation	3
Oral presentation at an international conference *Circle Meeting, IUPAB & EBSA, BIFI, MMEE,PLM,SFE2...*	15
Poster presentation at an international conference *StatPhys, PhysBio, AMR 2017, AMR 2018* – including 2 abroad	5
Seminars – including 10 invited seminars	15
Presentation at a local conference	13
Summer schools and thematic schools – including 1 as a lecturer	4

Grants, Awards & Prizes

2024 - 2029 **Max Planck Partner Group** – Collaborating grant of the MPG – 100 000 €

2023 - 2028 **CENTURI fellowship** – Group starting package – 100 000 € + 3 PhD/postdoc positions

2019 **CRC 1182 Lynn Margulis Prize** – Funding for a one-month research stay – 5000 €

2015 - 2018 **Three-year teaching assistantship** – selected by Université Pierre et Marie Curie, Paris

2015 - 2018 **Three-year full PhD fellowship** – selected by École Doctorale Physique en Île-de- France

2014 **Alsace and Lorraine General Association (AGAL) scholarship** - 1000€

Research interests

My research interests revolve around the modeling of microbial communities, in particular those that live in associations with hosts, in ecosystems named microbiomes I am particularly interested in importing classical models of ecology and evolution to the field of microbiome studies, by including some constraints that are specific to living in symbiosis with a host. For example, I aim at characterizing the selection gradient that results from the intertwining of the microbes and their host life cycles, taking into account traits like the host finite lifespan, migration of microbes from the environment and between hosts, possible interactions with the immune system, or microbial inheritance. I am also interested in identifying the mechanisms that favor the maintenance of the microbial community diversity and function.

Keywords: mathematical modeling; microbiota; microbial communities; population dynamics; stochastic processes; ecology; evolution; selection gradient; life cycles; immunity; feeding behavior; metabolism

Teaching

2021 **Teaching assistantship for master students** - *CAU zu Kiel, Germany*
Tutorials and evaluations - Evolutionary game theory for biologists

2019 **Teaching assistantship for master students** - *Universität zu Lübeck, Germany*
Tutorials and evaluations - Evolutionary dynamics: Game theory

2015 - 2018 **Teaching assistantship for undergraduates** (192 h) - *UPMC, Paris, France*
Tutorials and oral interrogations in basic mechanics and thermodynamics
Practicals in electrokinetics and electromagnetism

2014 - 2015 **Physics lessons** - *Institut Bossuet, Paris, France*
Monthly oral examinations in “classes préparatoires” (undergraduate level)

PEER-REVIEWED PUBLICATIONS

1. V. M. Marquioni, A-C. Hofacker, J. V. Villavicencio, F. Bansept, Modulating microbial intake helps to maintain the gut microbiome diversity, Communications Biology, 2026

2. R. Zapién-Campos, F. Bansept, A. Traulsen, Stochastic models allow improved inference of microbiome interactions from time series data, PLoS Biology, 2024

3. N. Obeng, A. Czerwinski, D. Schütz, J. Michels, J. Leipert, F. Bansept, M. J. García García, T. Schultheiß, M. Kemlein, J. Fuß, A. Tholey, A. Traulsen, H. Sondermann & H. Schulenburg, Bacterial c-di-GMP has a key role in establishing host–microbe symbiosis, Nature Microbiology, 2023

4. D. Hoces, G. Greter, M. Arnoldini, M.L. Stäubli, C. Moresi, A. Sintsova, S. Berent, I. Kolinko, F. Bansept, A. Woller, J. Häfliger, E. Martens, W-D. Hardt, S. Sunagawa, C. Loverdo, E. Slack, Fitness advantage of *Bacteroides thetaiotaomicron* capsular polysaccharide in the mouse gut depends on the resident microbiota, eLife, 2023

5. R. Zapién-Campos, F. Bansept, M. Sieber, and A. Traulsen, On the effect of inheritance of microbes in commensal microbiomes, BMC Ecol Evo, 2022

6. F. Bansept, N. Obeng, H. Schulenburg, and A. Traulsen, Modeling host-associating microbes under selection, The ISME Journal, 2021

7. N. Obeng, F. Bansept, M. Sieber, A. Traulsen, H. Schulenburg, Evolution of microbiota-host associations: the microbe's perspective, Trends in Microbiology, 2021

8. F. Bansept, L. Marrec, A-F. Bitbol and C. Loverdo, Antibody-mediated cross-linking of gut bacteria hinders the spread of antibiotic resistance, Evolution, 2019 (arXiv:1903.05723v1)

9. F. Bansept, K. Moor, M. Diard, W-D Hardt, E. Slack and C. Loverdo, Enchained growth and cluster dislocation: a possible mechanism for microbiota homeostasis, PLoS Comput Biol, 2019

10. K. Moor, M. Diard, M. E. Sellin, B. Felmy, S. Y. Wotzka, A. Toska, E. Bakkeren, M. Arnoldini, F. Bansept, A. Dal Co, T. Völler, A. Minola, B. Fernandez-Rodriguez, G. Agatic, S. Barbieri, L. Piccoli, C. Casiraghi, D. Corti, A. Lanzavecchia, R. R. Regoes, C. Loverdo, R. Stocker, D. R. Brumley, WD Hardt, E. Slack, High-avidity IgA protects the intestine by enchaining growing bacteria, Nature, 2017

PREPRINTS AND SUBMITTED MANUSCRIPTS

- F. J. F. Labourel, F. Bansept, D. M. McCandlish, Weakest link epistasis and the geometry of genetic load, (preprint on bioRxiv)

- A. Galeev, F. Bansept, N. Benny, V. Chaves Vargas, A. Habich, A. Traulsen, D. Unterweger, Type VI secretion systems of *Pseudomonas aeruginosa* modulate infection dynamics and immune responses in *Galleria mellonella* larvae, *in revision*

MANUSCRIPTS IN PREPARATION

- F. Bansept, L. Espinosa, C. Carrère, Evolution of microbial biphasic life cycles, *in preparation*

Supervision & Mentoring

Postdoc

2025 - 2026 **Advising of Alessandra Lütz** - CAPES-COFECUB fellow – *Aix-Marseille University, France*
Postdoc (1 year) - An evolutionary graph theory framework for host-associated microbiomes

2024 - 2026 **Advising of Vitor Marquioni-Monteiro** - *Aix-Marseille University, France*
Postdoc (2 years) - Modeling microbial dynamics resulting from intermittent feeding

2024 - 2026 **Advising of Florian Labourel** - *Aix-Marseille University, France*
Postdoc (2 years) - Metabolism and multi-stable equilibria

PhD

2024 - 2027 **Co-supervision (20%) of Matthew Jones' PhD** - *LCB & Aix-Marseille University, France*
Co-supervisor: Maxence Vincent - Salmonella HOCl stress response & resulting population dynamics

Interns

2026 **Supervision of Julien Vogelweid** - *AMU, France*
M2 internship (6 months) - Intermittent feeding & GLV dynamics

2026 **Co-supervision of Abdelli Benazzouz & Pierre Burnier** - *AMU, France*
BTS (2nd year) internship (6 weeks) - Building online applications for the group

2025 **Co-supervision of Sherine Touchene & Myriam Bensalem** - *AMU, France*
BTS (1st year) internship (6 weeks) - Building an html website for the group

2025 **Supervision of Moez Gouiaa** - *Aix-Marseille University, France*
M2 internship (6 months) - Using phages to control the gut microbiome: a mathematical model

2025 **Supervision of Valentine Baud** - *Aix-Marseille University, France*
M1 internship (3 months) - Metabolic fitness landscapes in fluctuating environments

2024 **Supervision of Dorothée Bertin** - *Aix-Marseille University, France*
M1 internship (3 months) - Biphasic life cycles with mutations

2024 **Supervision of Miguel Pedraza** - *Aix-Marseille University, France*
M2 internship (3 months) - tRNA genes evolution

2023 **Supervision of Jocksan Villaviciencio-Villavicencio** - *Aix-Marseille University, France*
M1 internship (5 months) - Modeling microbial dynamics resulting from intermittent feeding

2022 **Supervision of Ann-Cathrin Hofacker** - *Christian-Albrechts-Universität zu Kiel, Germany*
M1 (6 weeks) & Student helper contract (160h) financed by the program via:mento
Bibliographic project and simulations investigating the link between gut microbiota and feeding

2020 - 2021 **Supervision of Carla Gensana Claus** - *Christian-Albrechts-Universität zu Kiel, Germany*
Undergraduate internship, 2 months (2020) and Bachelor thesis (2021)
Epidemiological modeling around the COVID-19 pandemic in Spain

2020 **Co-supervision of Andonis Gerardos** - *Sorbonne Université, Paris, France*
Master internship, 4 months, co-advised with Claude Loverdo
Development of infering methods applied to the modeling of bacterial infections

Thesis committee

2024 - 2027 **Thesis committee (tutor) of Marie-Jose Chaaya** - *Aix-Marseille University, France*
Supervisor: Florence Hubert - Mathematical modeling of cancer/nervous system interactions

2024 - 2027 **Thesis committee (invited) of Florian Saby** - *LCB, Aix-Marseille University, France*
Supervisor: Tâm Mignot & Aurore Loisy - Understanding Decision-Making in *Myxococcus xanthus*

CONFERENCES & WORKSHOPS

July 21-26, 2026	**Mathematical Models in Ecology & Evolution** - *Cork University, UK* - contributed talk
Sept 18-19, 2025	**International Course on Multiscale Integration** - *Institut Curie, France* - invited talk
Sept 18-19, 2025	**Symposium "Systems Bio of Microbial Infections** - *Jena, Germany* - invited talk
July 28-31, 2025	**Partner Groups Workshop** - *Plön, Germany* - talk
March 21, 2025	**MathBio Day** - *Marseille, France* - invited talk
Nov 19, 2024	**IM2B Scientific Day** - *Marseille, France* - invited talk
Oct 21-25, 2024	**Congress of the French Society for Eco & Evo** - *Lyon, France* - talk
Sept 16-17, 2024	**Physics of Living Matter** - *Marseille, France* - invited talk
July 9-12, 2024	**Workshop "Math mod of microbial communities"** - *Leeds, UK* - invited talk
June 17-19, 2024	**Workshop "Developments in evolutionary Theory"** - *Plön, Germany* - talk
May 20-24, 2024	**Conference "Stochastic Processes in Evol Bio"** - *CIRM, France* - talk
Jan 9, 2024	**μNet day** - *Marseille, France* - talk
Oct 4-6, 2023	**Horizons in Metaorganism Research** - *Kiel, Germany* - talk
May 11, 2023	**CENTURI day** - *Marseille, France* - talk
April 17-21, 2023	**BEvAS workshop** - *Bernouilli center, EPFL, Switzerland* - invited talk
Oct 24-26, 2022	**CRC 1182 Retreat** - *Schleswig, Germany* - poster
Sept 5-7, 2022	**Workshop "Phys & chem det. of bio. evo."** - *Plön, Germany* - invited talk
July 18-20, 2022	**Mathematical Models in Ecology and Evolution** - *Reading, UK* - talk
March 21-25, 2022	**Workshop "Math. mod. in eco-evo"** - *Paris, France* - invited talk
March 10-12, 2022	**Workshop "Role of microbiomes"** - *Luzern, Switzerland* - invited talk
Nov 8-9, 2021	**Workshop Mathématiques & Microbiote** - *Besançon, France* - invited talk
Nov 4-5, 2021	**CRC 1182 Retreat** - *Timmendorfer Strand, Germany* - talk
June 17-18, 2021	**Aquavit - Internal symposium** - online edition - *MPI Plön, Germany* - poster
June 1-12, 2021	**Summer School: Stat Mech & Emergent Phenomena** - *Beg Rohu, France* - poster
Oct 26, 2020	**CRC1182 Retreat** - *Online* - short talk
June 19-23, 2020	[Cancelled] **Evolution 2020 (ASN, SSE, SSB)** - *Cleveland, Ohio, USA* - talk
June 11, 2020	**Aquavit - Internal symposium** - online edition - *MPI Plön, Germany* - talk
Aug 8-9, 2019	**Young Investigator Research Day of CRC 1182** - *Kiel, Germany* - poster
July 16-19, 2019	**Mathematical Models in Ecology and Evolution** - *Lyon, France* - talk
Nov 23, 2018	**Rencontre des Jeunes Physiciens** - *Paris, France* - talk
March 19-21, 2018	**AMR 2018** - *Institut Pasteur, Paris, France* - poster
Feb 6-8, 2018	**BIFI: Complexity, networks & coll. behaviour** - *Zaragoza, Spain* - short talk
Jul 24-Au 5, 2017	**Summer School: Out-of-Eq Dyn, Evo & Gen** - *Beg Rohu,, France* - poster
July 20-21, 2017	**AMR 2017** - *Edinburgh, UK* - poster
July 16-20, 2017	**IUPAB & EBSA congress** - *Edinburgh, UK* - talk
June 19, 2017	**Journée des Doctorants 2017 - EDPIF** - *Paris, France* - poster
March 8-10, 2017	**Physical Biology Circle Meeting** - *Heidelberg, Germany* - talk
Nov 3-9, 2016	**Course "Multiscale integration in Bio Sys"** - *Institut Curie, Paris, France* - talk
Oct 24-26, 2016	**Physics and Biological Systems 2016** - *École Polytechnique, France* - poster
July 18-22, 2016	**Statphys26** - *Lyon, France* - poster
Jan 28-29, 2016	**Days of statistical physics** - *Paris, France* - short talk
Nov 10, 2015	**Rencontre des Jeunes Physiciens** - *Paris, France* - poster

Seminars

November 24, 2025	**Invited webinar for the Biophysics Brazilian Society** - *online*
June 11, 2025	**Invited seminar at Centre de Recherche en Cancérologie** - *Marseille, France*
April 25, 2025	**Invited seminar at Institut Denis Poisson** - *Orléans, France*
October 1, 2024	**Invited seminar at the Institute of Ecology and Evolution** - *Bern, Switzerland*
September 26, 2024	**Invited seminar at Computational Biology Dept., Pasteur** - *Paris, France*
February 26, 2024	**Invited seminar at MABioS (I2M) & SB (MMG)** - *Marseille, France*
June 29, 2023	**Internal seminar at LCB** - *Marseille, France*
June 19, 2023	**Internal seminar at CENTURI** - *Marseille, France*
April 26-28, 2023	**Invited seminar at Inria/INRAE, Pléiade team** - *Bordeaux, France*
April 4, 2023	**Invited seminar (Sammba) at Institut Pasteur** - *Paris, France*
July 8, 2022	**Invited seminar at the Leibniz Institute on Aging** - *Jena, Germany*
June 2, 2022	**Invited seminar at the Turing Center for living systems** - *Marseille, France*
November 18, 2021	**Invited seminar at the LBBE lab** - *Lyon, France*
November 17, 2021	**Invited seminar at the Ecobio lab** - *Rennes, France*
September 29, 2020	**CRC1182 - Metaorganism Seminar** - *Online* - talk
April 25, 2018	**Invited seminar at the MPI for Evolutionary Biology** - *Plön, Germany*
May 24, 2016	**LJP internal seminar** - *Paris, France* - talk
June 17, 2015	**Institute for integrative biology seminar** - *ETH Zürich, Switzerland* - short talk

Languages

French	Native
English	Fluent
Portuguese	Good understanding and expression
German	Good understanding, basic conversation

Computer skills

Operating Systems	proficient user of Mac OS X, working knowledge of Linux and Windows
Programming	proficient user of R, working knowledge of Python and C
Scientific software	proficient user of Mathematica, working knowledge of Maple, Vision Assistant, Image J, Labview, git
Typesetting	proficient user of LaTeX and Pack Office

ACADEMIC SERVICES

Oct. 2026 **Co-organizer of the CENTURI Scientific conference** - *Cargèse, France*

May. 2026 **Co-organizer of a thematic day for Modeling in Eco-Evo** - *Marseille, France*

2025 **Jury of the annual CENTURI PhD Call** - *Aix-Marseille University, France*

Dec. 6, 2024 **Thesis Jury (examiner) of Mario Desallais** - *Toulouse III University, Moulis,France*
Supervisors: Jean-François Arnoldi & Michel Loreau
Rôle de la diversité et des interactions biotiques dans la stabilité des écosystèmes

Since 2024 **Equal Opportunity Officer & Communication Team**
Laboratoire de Chimie Bactérienne, *Marseille, France*

September 2024 **Co-organizer of the international conference Physics of Living Matter 18**
Palais du Pharo, *Marseille, France*

September 2022 **Co-organizer of the workshop Mathematical Modelling of Microbiomes**
Max Planck Institute for Evolutionary Biology, *Plön, Germany*

July 2022 **Co-organizer of a minisymposium within the MMEE conference**
"Cancer vs microbial community modeling: divides and bridges", *Reading, UK*

2020 **Outreach project on the Corona crisis** - with the MPI Plön theory department
Collaborative creation of an educational website for the general public: writing and translation of scientific articles about the pandemic, creation of simulation activities for highschool students.
One of them has been funded by the Robert Bosch Stiftung and produced as a real board game & distributed in over 10 000 copies over more than 5 countries

Since 2019 **Reviewer for Nature Com., Com. Bio, PCI Ecology, ISF grants,...**

2019 - 2022 **Member of the local Max Planck Sustainability Network**
Organizer of a Climate Emergency Journal Club and of related seminars

Nov 10, 2017 **Principal organizer of the Paris Biological Physics Community Day 2017**
A conference bringing into contact young scientists from the Paris region with invited external speakers

Oct 21, 2016 **Co-organizer of the Paris Biological Physics Community Day 2016**

FURTHER QUALIFICATIONS & CAREER DEVELOPMENT

2025 **Musical Studies Certificate (CEM)** - Major in Opera Singing - *CRR of Marseille, France*

2020 - 2022 **Mentee at Mentoring program via:mento** - *Selected by Kiel University, Germany*

2020 - 2022 **Coaching on Professional Positioning by Ute Jülly**

2019 - 2026 **Qualified for assistant professorship applications in France** (Sections 28, 29, 30, 67)

2018 **Musical Studies Diploma (DEM)** - Major in piano - *Conservatoire of Cachan, France*

2017 **Basic first aid skills certificate** (PSC1)

2011 **Driving Licence** (Permis B)

2010 **Musical Studies Certificate (CEM)** - Major in piano - *Conservatoire of Mulhouse, France*